11.4.2 Signals

Next, we will look at challenges related to the signals themselves.

Telemetry signals volume

The overall volume of data generated from different sources makes it difficult to correlate signals efficiently. This is not just a cost issue but can have serious impact on performance and flow. For example, if you're indexing logs and your tooling is overwhelmed serving user lookup requests, operators may be slowed down, frustrated, and effectively blocked.

The following are appropriate remediations:

- Employ selective ingest, meaning you limit the volume up front. This can be quite tricky, since you don't know ahead of time what the "valuable" signals are that help getting to actionable insights. For logging, make sure you have a `priority` or `loglevel` attribute attached to your events. This lets you do things like *ingest all errors, 50% of all warnings, and 1% of informational log lines*.
- An a posteriori approach refers to employing machine learning (ML) to detect patterns. This can help you deal with the volume, in that it enables you to isolate and prioritize relevant events that could lead to the desired insights. Be aware, though, that with ML-driven anomaly detection, the cost is in the compute, not storage, so the ingestion costs are far higher than signal retention.

High cardinality

High-cardinality data, as discussed in chapter 5 (especially the cardinality explosion challenge), can make it difficult to correlate signals. This is because values may only occur a few times or, in the most extreme case, once.

The following are appropriate remediations:

- Limit cardinality. This entails enforcing hard limits, with the downside being that you don't know, in the general case, what the right cutoff point is, and you, again, may lose out on valuable signals.
- Make sure the observability offering of your choice supports high cardinality, which is especially important for Honeycomb and Lightstep.

Sampling

In the case of distributed tracing, sampling (see chapter 8, where we discussed this in detail) can be a challenge for correlation. Missing spans can cause holes in the correlation path, and with that, the overall usefulness decreases. There are a few vendors that can help you with that; for example, Honeycomb has a smart sampler, which uses a different sample rate for errors than the happy path. This is a variation on pretagging valuable signals.

Data privacy and security

Many industries as well as geographies, such as Europe's General Data Protection Regulation (GDPR; https://commission.europa.eu/law/law-topic/data-protection_en), have policies requiring the de-identifying of data, such as removing personally identifiable